Firdose Anjum

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.39
Base PU College	Class XII (KSEAB (Karnataka School Examination and Assessment Board))	2023	97.17%
Presentation Public School	Class X (CBSE (Central Board of Secondary Education))	2021	95.40%

SCHOLASTIC ACHIEVEMENTS

- Secured **9.8/10 SGPA** in 2 semesters, maintaining 9+ SGPA across all semesters at IIT Hyderabad.
- Received the **IIChE Meritorious Performance Award** for 2025-26.

POSITIONS OF RESPONSIBILITY

Finance Team, Department Day, IITH (January 2026 - February 2026)

- Handled fundraising and financial coordination, contributing to nearly 2 lakh collected for the department event.

Secretary, IIChE Student Chapter, IITH (October 2025 - Present)

- Organized multiple industry visits, coordinating communication and logistics with companies and participating students.

Multimedia Coordinator, International Relations Cell - IITH (May 2024 - May 2025)

- Designed high-impact visual **content** and managed digital communication.

- Curated event videos and social media visuals that amplified global outreach efforts by **2**x engagement.

EXPERIENCE

Amgen, GRAET Intern (May 2026 - July 2026)

- Developed an automation tool for PDF bookmarking, saving ~2 hours per person and improving workflow speed by 3×.
- Built a Power BI dashboard with interactive drill-downs to track projects across Amgen's regulatory domain.
- Designed a system architecture map in Visio to document data flows and application integrations across multiple teams.

SKILLS

- **Programming:** C, C++, Python, SQL
- **ML & Data Science:** Supervised & Unsupervised Learning, Deep Learning, Computer Vision, EDA, Time Series Analysis
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Statsmodels
- **Tools:** Power BI, Streamlit, Git, GitHub, MATLAB, Aspen Plus, Solid Edge
- **DSA:** Data Structures & Algorithms; 250+ problems solved on LeetCode

RELEVANT COURSES

- ID2230 - Data Structures & Applications
- ID1050 - Artificial Intelligence
- CH5720 - AI/ML Tool Chains for Chemical Engineers
- CH5483 - Computer Vision in Chemical & Life Sciences
- CH5010 - Advanced Numerical Methods
- CH5150 - Optimization Techniques-I

PROJECTS

ML Interatomic Potential for Fe-Cr System

- Compiled **~1,000 DFT-derived** atomic configurations for the Fe-Cr system to build a dataset for MLIP development.
- Reviewed literature to select Deep Potential (DP) and Behler-Parrinello Neural Network (BPNN) models for the system.
- Trained **DP and BPNN models** on the DFT dataset to predict atomic energies and forces.

Network Intrusion Detection using RBM (Restricted Boltzmann Machine)

- Implemented an RBM-based pipeline to learn latent representations from network traffic data.
- Combined RBM features with SVM to classify network traffic as normal or anomalous.
- Achieved **88.2% test accuracy and 0.88 macro F1-score** on binary intrusion detection.

Mass Transfer Surrogate Modeling & Optimization

- Developed a multistage solid-liquid extraction simulator for crosscurrent and countercurrent operations using nonlinear equation solving (fsolve) and experimental equilibrium data.
- Trained an **ANN (MLPRegressor) surrogate model** on synthetic data to predict oil removal based on stage count, solvent amount, feed concentration, and extraction mode.
- Built heatmaps, contour plots, and 3D response surfaces and implemented **surrogate-assisted optimization** to identify efficient solvent-stage combinations.
- Results: Achieved **99.40% oil removal** in countercurrent and 96.88% in crosscurrent for three-stage extraction; reported approximately 98% agreement between graphical, numerical, and ANN results.

EXTRACURRICULAR

- Avid reader; read *The 7 Habits of Highly Effective People*, *Ikigai*, *Show Your Work!* and others.
- Enjoy playing badminton and chess.
- Amateur video editor; create and edit short-form content as a hobby.